

# Osaid Khan

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Dear Hiring Team,

It is with great pleasure that I submit my candidacy for the **Backend Developer** position with **Allstate**. I bring over three years of professional experience in building scalable backend systems, RAG pipelines, and AI agents, along with a strong academic foundation as a Master of Science in Computer Science graduate from Pace University.

Through my experience across engineering roles, I have built and delivered production-grade backend services that directly reflect **the backend scalability and AI integration depth this position demands**. As a Software Engineer at Sperse, I architected a multi-agent orchestration system using **LangChain** and **LangGraph**, enabling scalable **asynchronous processing** and reliable coordination across complex customer workflows. To ensure operational excellence, I developed an MCP AI interaction layer that translates natural-language intents into validated tool calls across 1,000+ microservices endpoints, and optimized agent concurrency to reduce end-to-end latency by 60%. Prior to Sperse, as a Software Engineer at Qualitest, I contributed to the backbone infrastructure that thousands of users rely on daily. I designed and built an internal support chatbot using **LangChain** and **RAG** with a **PostgreSQL vector store**, reducing ticket resolution time by 20%. I also contributed to an Identity & Access Management platform using **Java Spring Boot**, **React**, and **AWS**, supporting thousands of users across 30+ applications within a microservices architecture. These roles have prepared me to **contribute meaningfully to Allstate's engineering team by building secure, high-performance microservices and integrating cutting-edge AI capabilities**.

My academic research experience has given me a rigorous foundation in model development and technical communication — skills that translate directly into **the engineering excellence and collaborative culture Allstate values**. As an AI Researcher at the Pace Artificial Intelligence Lab, I conducted **fine-tuning** and optimization of **Large Language Models (LLMs)** and diffusion models using **PyTorch** and **LORA** on an HPC cluster. I was responsible for managing the full model lifecycle — from training to high-performance inference using **vLLM** and **Hugging Face** — ensuring both performance and secure model serving. Beyond technical implementation, I led workshops on building AI Agents with LangChain for an audience of 50+ students, which required breaking down complex technical concepts clearly. This experience has shaped me into an engineer who builds with rigor and communicates with clarity, both of which I would bring to **Allstate's mission of using innovation to protect customers**.

I have always been passionate about **excellence and being a force for good** and envision myself contributing my talent to the furthering of **Allstate's goal to help people protect what they have today and better prepare for tomorrow**. I would love to discuss my interest in this position with your team and can be reached at +1 (945) 304-5781 or [khanosaid726@gmail.com](mailto:khanosaid726@gmail.com). Thank you for your time and consideration, I look forward to connecting!

Sincerely,  
Osaid Khan